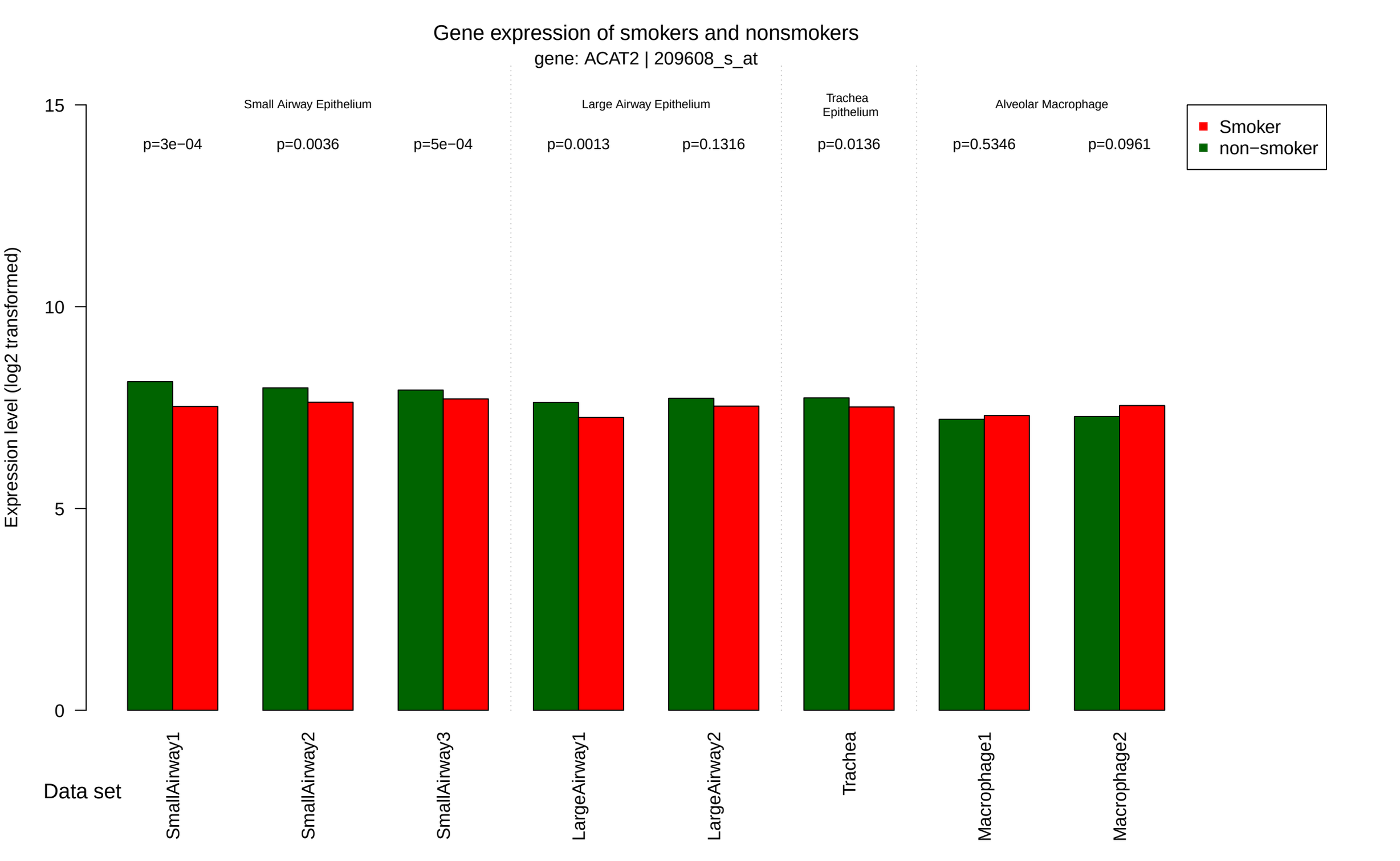
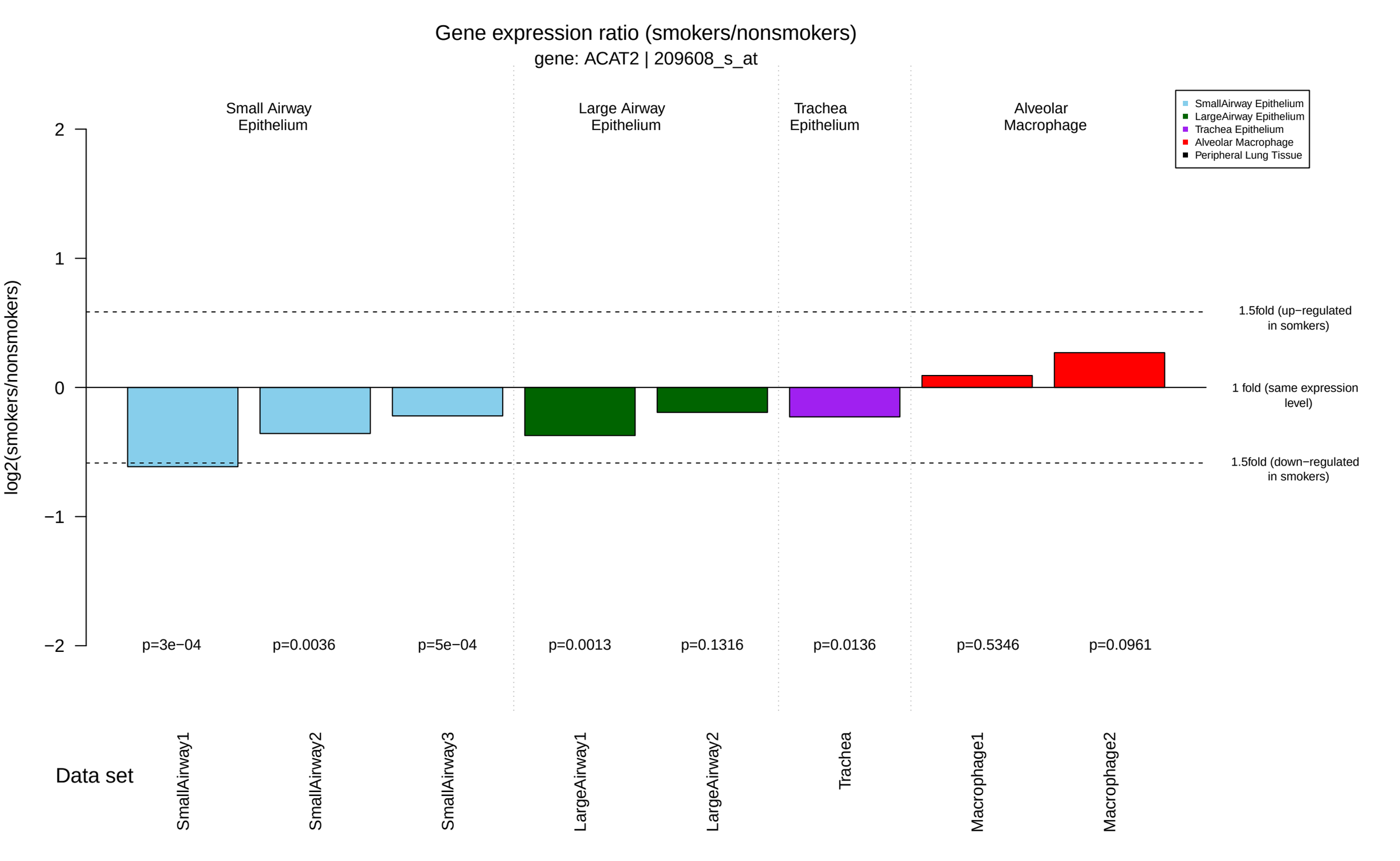
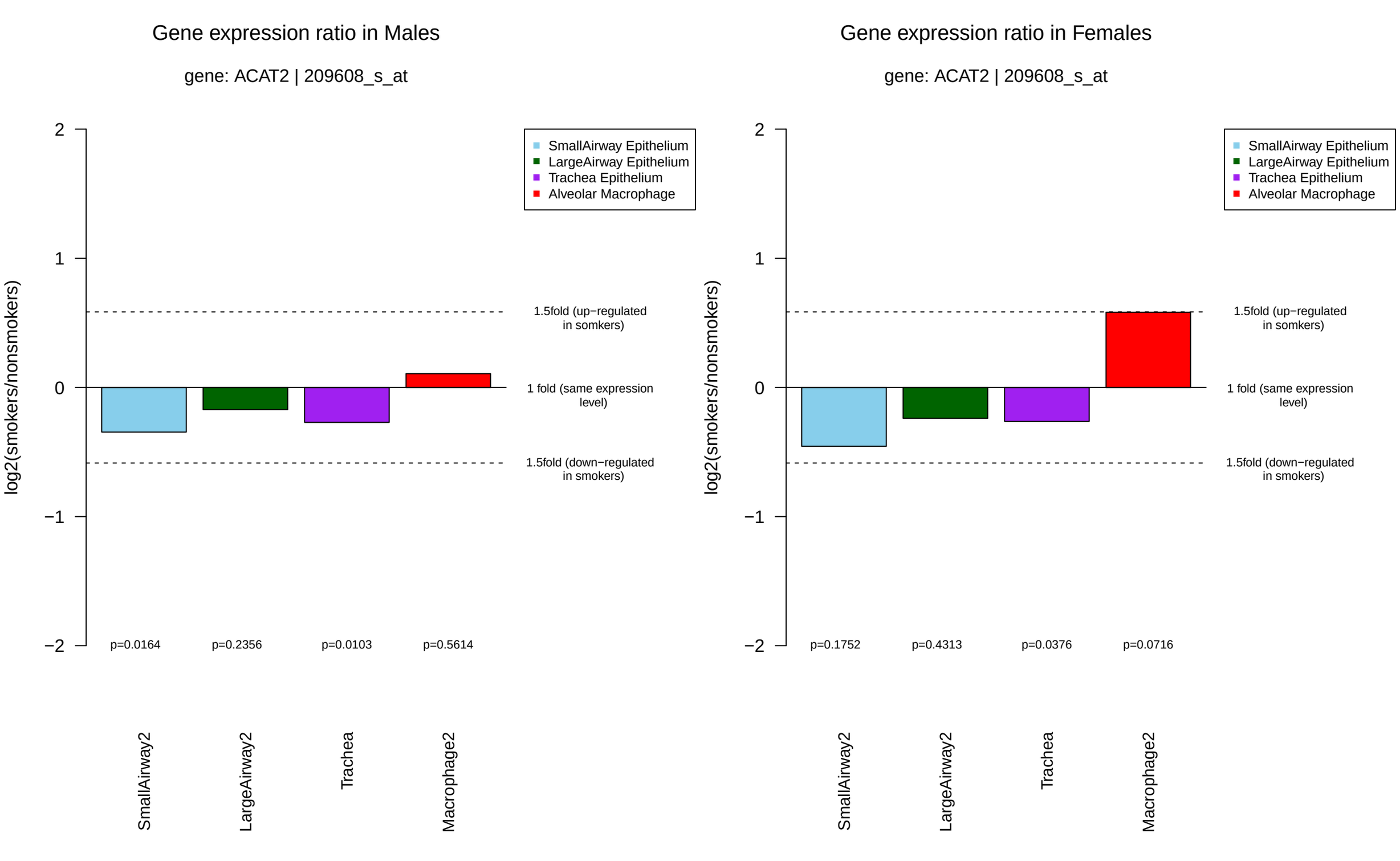
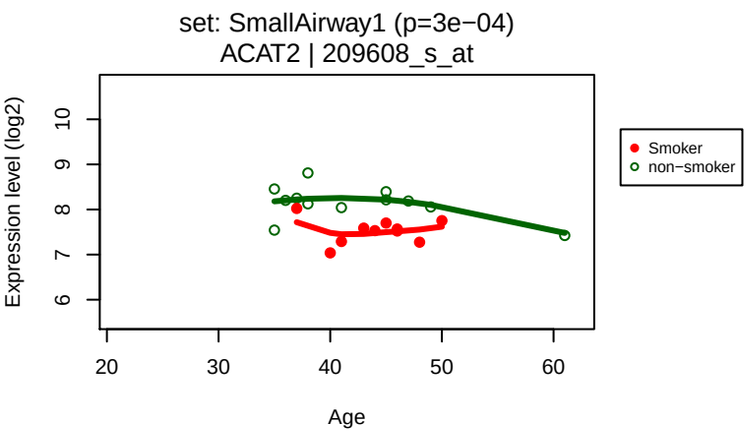
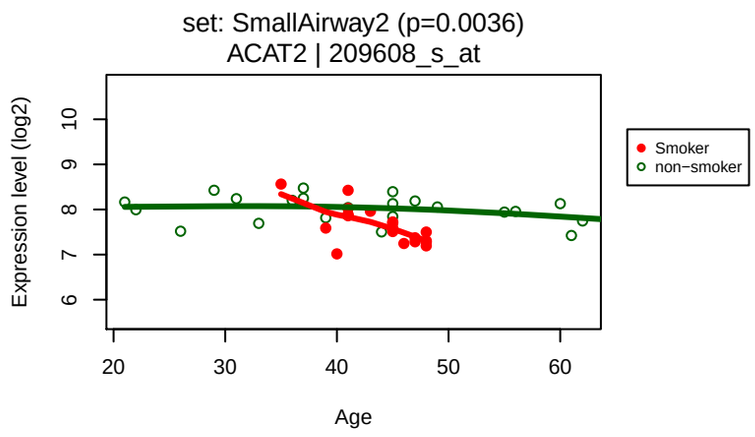
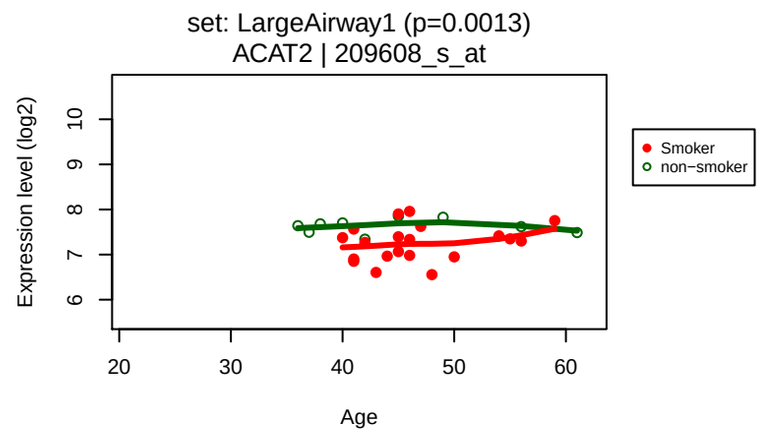

Gene expression of smokers and nonsmokers
gene: ACAT2 | 209608_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=3e−04
p=0.0036
p=5e−04
p=0.0013
p=0.1316
p=0.0136
p=0.5346
p=0.0961
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

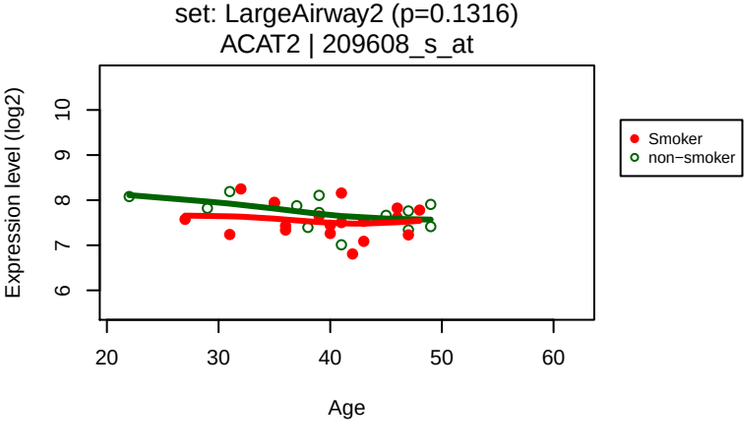

Gene expression ratio (smokers/nonsmokers)
gene: ACAT2 | 209608_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=3e−04
p=0.0036
p=5e−04
p=0.0013
p=0.1316
p=0.0136
p=0.5346
p=0.0961
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

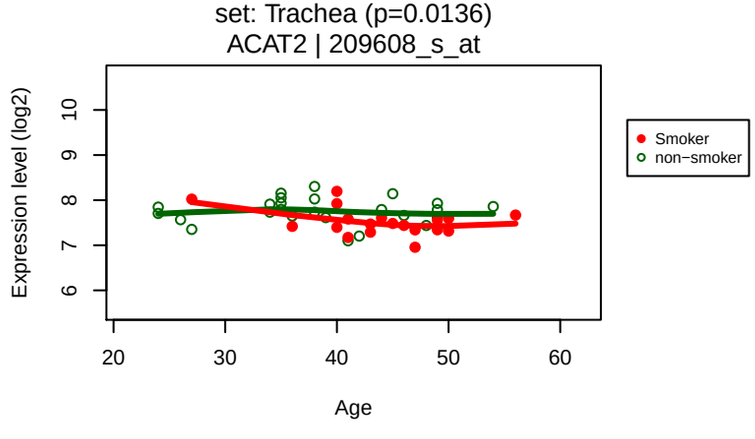

Gene expression ratio in Males
gene: ACAT2 | 209608_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0164
p=0.2356
p=0.0103
p=0.5614
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAT2 | 209608_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1752
p=0.4313
p=0.0376
p=0.0716
SmallAirway2
LargeAirway2
Trachea
Macrophage2

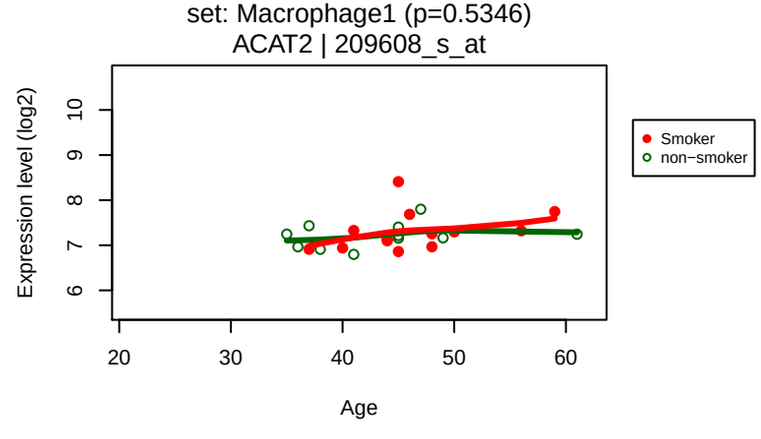

set: SmallAirway1 (p=3e−04)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker

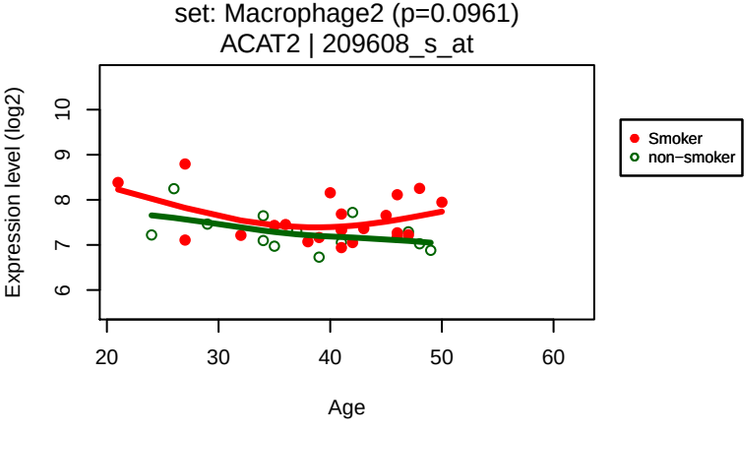

set: SmallAirway2 (p=0.0036)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker

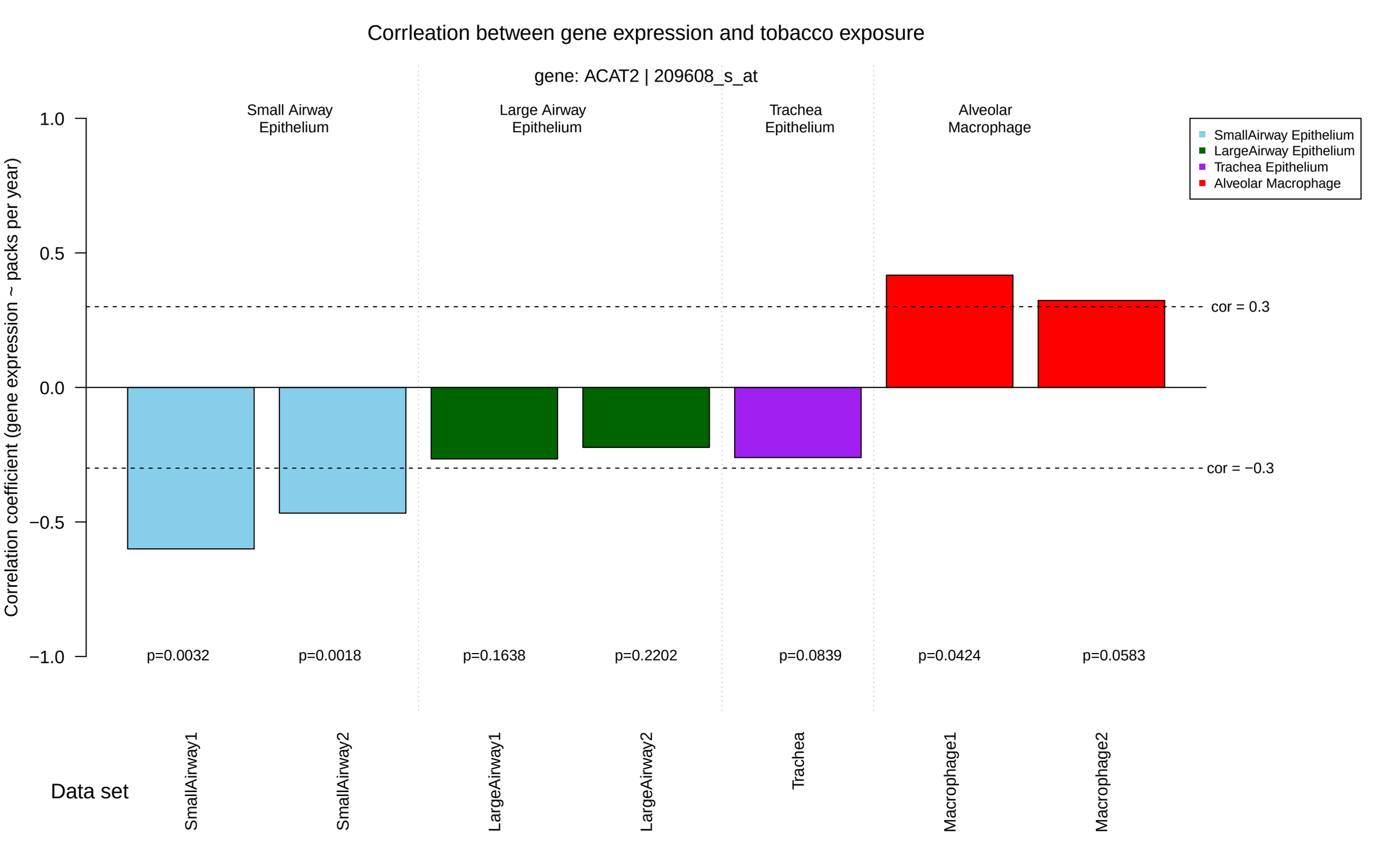

set: LargeAirway1 (p=0.0013)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1316)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0136)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5346)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0961)
ACAT2 | 209608_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAT2 | 209608_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0032
p=0.0018
p=0.1638
p=0.2202
p=0.0839
p=0.0424
p=0.0583
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2